Module 6: Final Data Science Project

Complete End-to-End Implementation Script & Source Code

Project: End-to-End ML Pipeline | Status: Complete Final Code | Milestone: Graduation Ready

1. Complete End-to-End Python Script

Below is the final, production-ready Python script containing the complete workflow for ingestion, preprocessing, training, evaluation, and serialization for Module 6.

```
# ========================================== # MODULE 6: FINAL DATA SCIENCE PROJECT SCRIPT # ========================================== import pandas as pd import numpy as np import matplotlib.pyplot as plt import seaborn as sns from sklearn.model_selection import train_test_split from sklearn.preprocessing import StandardScaler from sklearn.ensemble import GradientBoostingClassifier from sklearn.metrics import accuracy_score, classification_report, confusion_matrix import joblib # 1. Data Ingestion & Loading print("[INFO] Loading dataset...") # Replace with your actual dataset filename df = pd.read_csv('../data/dataset.csv') # 2. Data Cleaning & Preprocessing print("[INFO] Preprocessing data...") # Handling missing values df.fillna(df.median(numeric_only=True), inplace=True) # Feature & Target Separation X = df.drop(columns=['target']) y = df['target'] # Train-Test Split X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random_state=42, stratify=y ) # Feature Scaling scaler = StandardScaler() X_train_scaled = scaler.fit_transform(X_train) X_test_scaled = scaler.transform(X_test) # 3. Model Training (Gradient Boosting Classifier) print("[INFO] Training Gradient Boosting model...") model = GradientBoostingClassifier( n_estimators=150, learning_rate=0.1, max_depth=5, random_state=42 ) model.fit(X_train_scaled, y_train) # 4. Model Evaluation print("[INFO] Evaluating model performance...") predictions = model.predict(X_test_scaled) acc = accuracy_score(y_test, predictions) print(f"Model Accuracy: {acc * 100:.2f}%") print("\nClassification Report:") print(classification_report(y_test, predictions)) # 5. Save Artifacts print("[INFO] Saving trained model and scaler...") joblib.dump(model, 'final_model.pkl') joblib.dump(scaler, 'scaler.pkl') print("[INFO] Pipeline execution complete successfully!")
```

2. Complete GitHub `README.md` Template

Include this markdown text inside your repository's root directory to fulfill all documentation requirements:

```
# Final Data Science Project - Module 6 ## 📌 Project Overview This repository contains the complete end-to-end implementation for Module 6. The project covers the full data science lifecycle, from exploratory data analysis and feature engineering to model training and performance tuning using advanced machine learning algorithms. ## 📂 Repository Structure - `data/`: Houses the raw and processed dataset files. - `notebooks/`: Contains the executable Google Colab Jupyter notebook. - `src/`: Python source scripts for data preprocessing and model training. - `requirements.txt`: Required dependencies for running the environment. ## 🚀 How to Run 1. Clone the repository: `git clone https://github.com/username/repo-name.git` 2. Install packages: `pip install -r requirements.txt` 3. Execute training script: `python src/train.py`
```